

SMART INDIA HACKATHON 2026



- **Problem Statement ID – SIH26043**
- **Problem Statement Title – AI based Elephant attack alert system software**
- **Theme – Smart Automation / Wildlife Conservation & Disaster Management**
- **PS Category – Software**
- **Team ID – [KEEP EXISTING VALUE / PLACEHOLDER]**
- **Team Name – The Outliers**
- **Project Title – ELEPHANT GUARD**
- **Subtitle – Smart Elephant Early Warning & Human-Wildlife Conflict Mitigation System**

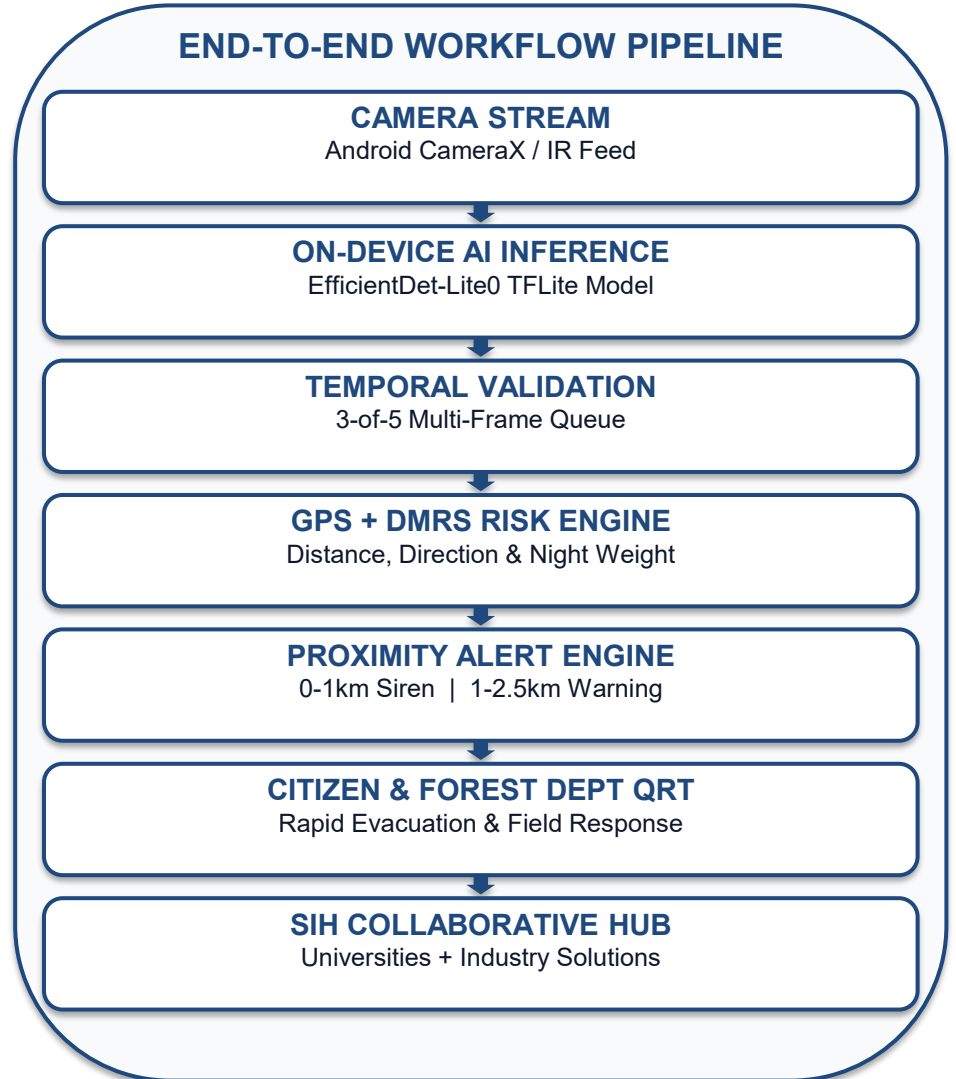
THE OUTLIERS



❖ Proposed Solution

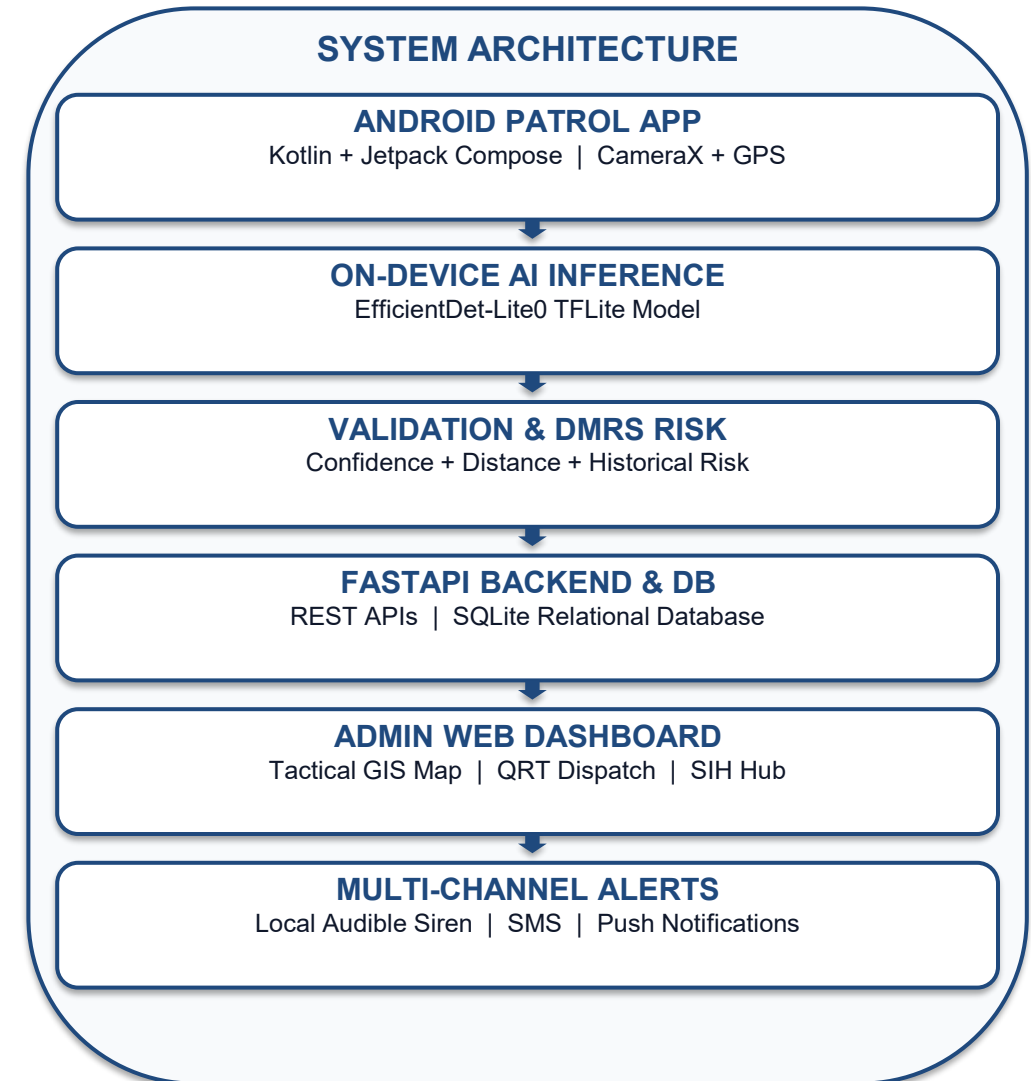
- **1. AI Early Warning:** On-device CV detects elephants with estimated GPS location.
- **2. Temporal Validation:** 3-of-5 frame queue validates detection to eliminate false alarms.
- **3. Dynamic Risk (DMRS):** Multi-factor score: proximity, confidence, direction & time.
- **4. Proximity Alerts:** 0–1 km: Siren + Critical alert | 1–2.5 km: Warning | >2.5 km: Safe.
- **5. Forest Dept Response:** Real-time GIS map, live incident feed & rapid QRT dispatch.
- **6. Collaborative SIH Hub:** Verified incidents converted to challenges for universities & industry.

END-TO-END WORKFLOW PIPELINE



Technology-

- **Frontend / Mobile:** Android Native (Kotlin, Jetpack Compose, CameraX, osmdroid).
- **AI / ML Pipeline:** TensorFlow Lite, EfficientDet-Lite0 on-device, Temporal Validation.
- **Backend Engine:** FastAPI (Python 3.11), REST APIs, Asynchronous WebSockets.
- **API / Sync Bus:** Telemetry intake, Incident sync, Offline queue via WorkManager.
- **Database:** SQLite relational schema: Users, Incidents, Alerts, Challenges, Logs.



Feasibility-

- **Technical Feasibility:**

On-device lightweight AI removes continuous cloud dependence. Phone Camera + GPS provide primary sensing.

- **Operational Feasibility:**

Designed for low-connectivity forest fringes. Multi-channel warnings (app, siren, SMS) ensure wide reach.

- **Economic Feasibility:**

Built on open-source frameworks. Uses commodity smartphones rather than costly military hardware.

Viability-

- **Social Viability:**

Early warnings reduce human encounters, protecting lives and crops. Connects communities directly with Forest Rangers.

- **Environmental Viability:**

Non-invasive camera detection causes zero harm to wildlife. Helps eliminate retaliatory killings and supports Project Elephant.

KEY CHALLENGES & ENGINEERED SOLUTIONS

- ⚠ **Detection Uncertainty**

→ Temporal validation (3-of-5 queue) + confidence filtering

- ⚠ **Camera Coverage Gaps**

→ Mobile sensing units + integration with public cameras

- ⚠ **Spam / Fake Reports**

→ Incident verification + AI media checking + trust scores

- ⚠ **Network Blackouts**

→ On-device edge AI + offline-first sync architecture

➤ Benefits-

- **Social Benefits:**

Faster early warning reduces human injuries & deaths. Protects crops and property. Multi-channel alerts (app, siren, SMS).

- **Technical Benefits:**

Demonstrates practical on-device edge AI. Combines AI, GPS, DMRS risk & GIS. Reusable for tigers & leopards.

- **Environmental Benefits:**

Non-invasive monitoring without disturbing elephants. Reduces retaliatory killing. Supports data-driven conservation.

➤ Potential Impact-

1. IMPACT & INTERVENTION PIPELINE



2. SIH COLLABORATIVE PROBLEM-SOLVING CYCLE



Research & References:

- **Down To Earth:** State of India's Environment 2025 (conflict data & trends).
- **PMC / PubMed:** Human-elephant conflict studies (Jharkhand & Chhattisgarh).
- **Nature (Reports):** Scientific research on HEC fatality trends & mitigation.
- **LILA BC:** Open wildlife camera-trap datasets.

Existing AI / Wildlife Platforms:

- **ElephantWatch:** Amrita University (AI-enhanced detection).
- **EleSense:** Voice for Asian Elephants (Sensor corridor alert).
- **MegaDetector:** Microsoft AI for Earth (Open-source wildlife model).

Competitive Analysis

Key Parameter	ElephantWatch	EleSense	Public Cams	ELEPHANT GUARD
Real-time detection	✓ (Server)	✓ (Sensor)	— (Manual)	✓ (On-Device AI)
Risk-zone assessment	— (Basic)	— (Track)	— (None)	✓ (DMRS Engine)
Temporal validation	— (Single)	— (Threshold)	— (None)	✓ (3-of-5 Queue)
Proximity alerting	— (General)	✓ (Local)	— (None)	✓ (0-1km / 1-2.5km)
Forest Dept dashboard	✓ (Monitor)	— (Hardware)	— (Feed only)	✓ (Full Web GIS)
Crowdsourced reports	— (Sensor)	— (Sensor)	— (None)	✓ (Citizen + QRT)
University / Industry hub	— (Internal)	— (NGO)	— (None)	✓ (SIH Challenge)
Scalable to other wildlife	— (Elephant)	— (Elephant)	✓ (Manual)	✓ (Modular)